BORA paper-data: figure/table review

January 1, 1970

Table 1: Evaluation datasets: Mal (malware scanning), Dna (genomic variants), and Dlp (e-mail DLP).

	Document corpus				Regex set		
	Corpus	Docs	Size	Per-doc (min/med/max)	Ruleset (ver.)	Rules	Size
Mal	CentOS 7	1,209	765 MB	66 KB / 130 KB / 33 MB	ClamAV 0.103.11	38,875	11.6 MB
Dna	chr17 (GRCh38)	1	83 MB	— (single)	NCBI ClinVar (chr17)	27,500	1.4 MB
Dlp	Enron Email	509,610	1.38 GB	398 B / 1.5 KB / 1.7 MB	MS-DLP (SIT)	136	213 KB

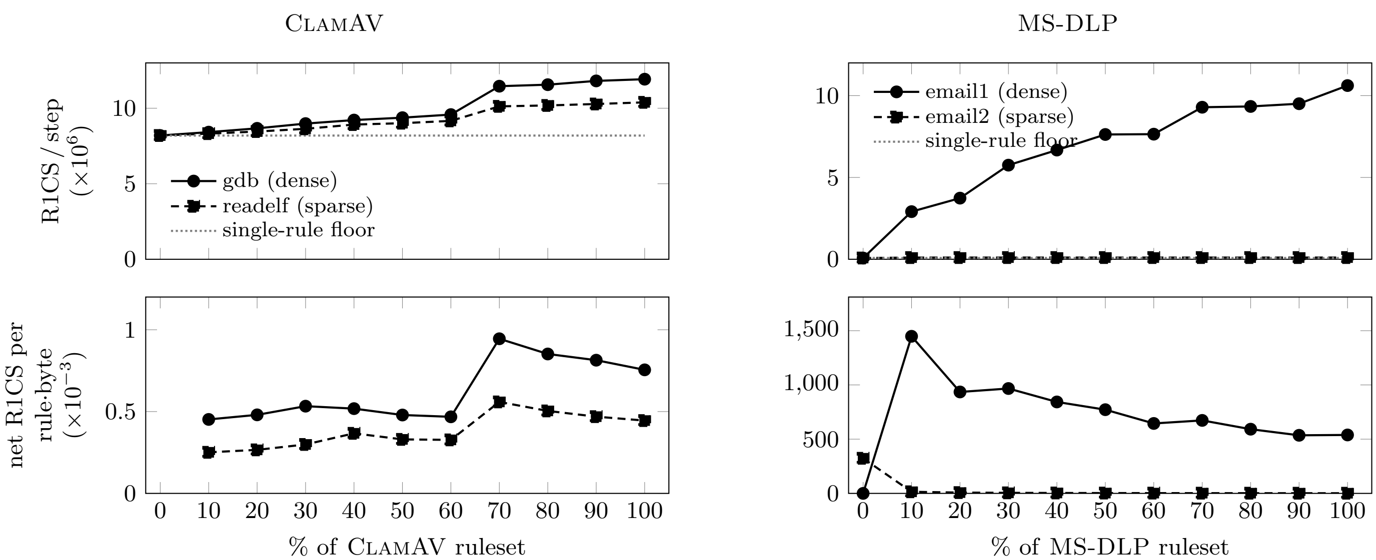

Figure 1: Ruleset-size scalability (input fixed), per folding step, on ClamAV (left) and MS-DLP (right). *Top:* absolute circuit size with the single-rule framework floor (dotted; 8.19 M R1CS for ClamAV, 0.08 M for MS-DLP); growing the ruleset to the full 38,875 (ClamAV) / 9,861 (MS-DLP) rules grows the circuit far slower than the rule count. *Bottom:* net circuit cost *per rule per input byte*—cumulative R1CS above the floor divided by cumulative rule count times per-step input length—moving from 0.45×10^{-3} to 0.76×10^{-3} (`gdb`) and 0.25×10^{-3} to 0.44×10^{-3} (`readelf`) R1CS per rule-byte (ClamAV) and 1448.30×10^{-3} to 538.78×10^{-3} (`email1`) and 14.82×10^{-3} to 1.53×10^{-3} (`email2`) R1CS per rule-byte (MS-DLP) between 10% of the ruleset and the full set. For the same MS-DLP ruleset, Zombie emits 77,463,552 R1CS in total; dividing by the 2000-byte input and the same 9,861 rules gives ∼3.93 R1CS per rule-byte, the same normalization applied to BORA.

Dataset	Pairs	`CP`	`SDE`	`DFA`
Mal	4.70×10^7	99.8803%	0.1144%	0.0053%
Dna	2.75×10^4	99.9382%	0.0618%	0.0000%
Dlp	4.98×10^9	99.9980%	0.0020%	0.0000%

Table 2: Tier-discharge fractions per dataset.

Size (B)	Files	`CP`	`SDE`	`DFA`
2^{16}–2^{17}	633	99.9225%	0.0723%	0.0052%
2^{17}–2^{18}	163	99.9069%	0.0879%	0.0052%
2^{18}–2^{19}	158	99.8872%	0.1076%	0.0052%
2^{19}–2^{25}	255	99.7544%	0.2399%	0.0057%

Table 3: Tier-discharge fractions by file size (Mal).

Size (B)	Files	`CP`	`SDE`	`DFA`
2^{7}–2^{10}	154012	99.9997%	0.0003%	0.0000%
2^{10}–2^{11}	172223	99.9987%	0.0013%	0.0000%
2^{11}–2^{12}	110432	99.9975%	0.0025%	0.0000%
2^{12}–2^{20}	68187	99.9933%	0.0067%	0.0000%

Table 4: Same, over Dlp.

Component	Purpose	Mal M	Mal %	Dna M	Dna %	Dlp M	Dlp %
range table†	range proof e.g. for pattern distance	67.1	27.3	134.2	81.6	33.6	28.2
AC-DFA / CP	transition rules	54.8	22.3	14.8	9.0	0.3	0.2
SDE	subsig keyword AC-DFA + step distance bounds	122.0	49.5	15.3	9.3	82.8	69.6
DFA (35/0/0)	transition rules	2.0	0.8	0.0	0.0	0.0	0.0
sig-DB	DNF combinations / tri-logic truth table	0.3	0.1	0.1	0.0	2.3	1.9
TOTAL populated		246.2		164.4		118.9	
#signatures		38,875		27,500		9,861	

Table 5: Lookup-table composition across datasets. Column M is millions of entries. Each % is of that dataset's populated entries. †range table = $2^{\text{range2\_bit}}$ (26 / 27 / 25 for Mal/Dna/Dlp), sized by the maximum document offset. The $(\cdot/\cdot/\cdot)$ after *DFA* is the number of DFAs folded in for rules that frequently fail SDE, per dataset. BORA re-writes `(kw1|...kwn).{0,N}pat` in Dlp to disjunction of n rules for improved SDE accuracy, thus expanding from 136 to 9861.

Dataset	Circ	input (B)	total R1CS	R1CS / input byte: raw (adjusted) $\mathsf{cost}_{\mathtt{CP}}$	$\mathsf{cost}_{\mathtt{SDE}}$	$\mathsf{cost}_{\mathtt{DFA}}$	$\mathsf{cost}_{\mathtt{FRWK}}$	corpus share
Mal	0	126,976	15,124,235	10.5 (32.1)	21.3 (65.1)	5.2 (5.5)	82.2 (16.3)	69.5%
	1	126,976	24,119,409	11.0 (30.3)	52.6 (134.9)	8.2 (8.5)	118.2 (16.3)	30.5%
Dna	0	126,976	13,913,921	28.7 (64.0)	22.4 (45.1)	0.0 (0.0)	58.5 (0.6)	100.0%
Dlp	0	1,984	138,606	5.5 (10.2)	23.5 (43.9)	0.0 (0.0)	40.9 (15.8)	92.6%
	1	1,984	1,513,906	5.5 (7.1)	571.4 (740.2)	0.0 (0.0)	186.2 (15.8)	6.5%
	2	1,984	6,631,976	5.6 (7.0)	2654.7 (3319.9)	0.0 (0.0)	682.4 (15.8)	0.8%
	3	1,984	17,189,459	5.6 (7.0)	6915.4 (8641.2)	0.0 (0.0)	1743.0 (15.8)	0.1%

Table 6: Per-circuit cost profile. Each dataset is discharged by a small set of circuits. *input* is the bytes processed per folding step. Each $\mathsf{cost}_t \in (\mathsf{cost}_{\mathtt{CP}}, \mathsf{cost}_{\mathtt{SDE}}, \mathsf{cost}_{\mathtt{DFA}}, \mathsf{cost}_{\mathtt{FRWK}})$ is the per-input-byte R1CS constraints of the four components of a circuit.Each cost_t is shown as *raw (adjusted)*: the *adjusted* value moves the *measured* log-up query cost in $\mathsf{cost}_{\mathtt{FRWK}}$ and charges it to each tier in proportion to its data-column size, so $\mathsf{cost}_{\mathtt{CP}}/\mathsf{cost}_{\mathtt{SDE}}/\mathsf{cost}_{\mathtt{DFA}}$ reflect the all-in per-byte cost and $\mathsf{cost}_{\mathtt{FRWK}}$ retains only folding overhead and the processing of Logup share of lookup table.

Dataset	Corpus	Jobs	Steps	Stage 1: Main Folding Sel.	Commit wit.	Fold	Total / Wall (jobs)	Stage 2 Compression
Mal	793.4 MB	8	6,552	12.74 hr	14.61 hr	89.64 hr	116.99 hr / 15.48 hr (8)	2.84 hr
Dna	39.7 MB	1	328	18.0 min	25.9 min	2.43 hr	3.16 hr (1)	1.28 hr
Dlp	1.7 GB	8	926,900	38.68 hr	43.24 hr	426.02 hr	507.94 hr / 65.60 hr (8)	1.76 hr
Per batch proof (one per job, constant): size 1,152 B, verification 32 ms.								

Table 7: End-to-end prover performance. *Corpus* is the bytes actually folded: Dna packs each base of its chr17 sequence into a nibble (half a byte), halving it to the 39.7 MB shown, while the 793.4 MB Mal results from padding each chunk to 128 KB. *Stage 1* (main folding) is the corpus-scaling cost: circuit selection + commit fixed witness + folding. *Total / Wall (jobs)* reports the summed CPU-time over the parallel jobs and the wall-clock time, i.e. the slowest single job, since the jobs run in parallel. *Stage 2: Compression* computes the main-circuit SNARK proof, the QA-NIZK proof, the cycle-pairing fold, and the resulting batch proof. It is a fixed per-job cost, and all timings exclude the one-time key setup. The bottom row shows the size and verification cost of one batch proof (mainly the two Groth16 SNARK proofs); this per-job batch proof is constant size and independent of corpus and rule-set size. A run emits one batch proof per job, as jobs split a corpus to save wall time.

Table 8: Reef per-bucket cost vs. BORA on chr17 $\times$ NCBI (27,500 regexes). Both systems are compared on the *net* step-folding cost only, excluding commitment, setup, and IVC proof compression, for a fair comparison. For Reef, net = witness generation + Nova prove + SAFA solve (`fa_solver`); it excludes the fixed floor (the 4.3 GB commitment load, SNARK setup, consistency proof, and related one-time costs). For BORA, net = the Phase 1 circuit selection and non-deterministic advice generation, and Phase 2 main-folding step cost, excluding the Groth16 IVC proof compression as well as setup and commitment. Estimate1 uses each bucket's measured per-run mean times its population count; Estimate2 uses the minimum measured per-run mean (proj_512k) times bucket count as a counterfactual floor. Each per-run mean is over $n = 10$ sampled runs; $\pm$ is their population s.d.

Bucket	Count	Per-run net cost (s)	Estimate1 (s)	Estimate2 (s)
non_projectable	24,434	1055.35 ± 68.70	2.579×10^7	1.791×10^5
proj_512k	58	7.33 ± 0.38	4.251×10^2	4.251×10^2
proj_2M	132	18.43 ± 0.90	2.433×10^3	9.676×10^2
proj_4M	1,194	35.04 ± 1.73	4.184×10^4	8.752×10^3
proj_8M	869	67.25 ± 4.15	5.844×10^4	6.370×10^3
proj_16M	813	137.35 ± 10.57	1.117×10^5	5.959×10^3
Reef TOTAL	**27,500**	—	2.600×10^7 (300.94 d)	2.016×10^5 (2.33 d)
BORA	**27,500**	— (single batched run)	1.138×10^4 (3.16 hr) — $\approx 2284\times$ faster	1.138×10^4 (3.16 hr) — $\approx 17.71\times$ faster

Table 9: Zombie prover cost on the MS-DLP international SIT rules (Spartan NIZK, two-pattern proximity circuit), totalled over all 194 regex instances (the 136 SIT rules of Table 1, expanded into their comb variants) at each scanned document length L_T. Zombie proves each instance separately; constraints, prove and verify time, and proof size are all summed over these instances. The total regex size is fixed at 213,232 B. The last row derives the unit cost $u = t_{\text{prove}}/(L_T \cdot \sum_{r \in \hat{R}} |r|)$ in seconds per (input-byte $\times$ regex-byte).

L_T (B)	Total constraints	Prove (s)	Verify (s)	Proof (MB)
1,000	38,731,776	654.8	336.4	6.16
2,000	77,463,552	2169.0	1281.9	7.48
4,000	154,927,104	8730.7	5618.9	9.90
Unit $u = t/(L_T \cdot \sum_{r \in \hat{R}} |r|)$ at $L_T = 1,000/2,000/4,000$: $3.0707 \times 10^{-6}/5.0861 \times 10^{-6}/1.0236 \times 10^{-5}$ s B^{-2}				

Table 10: Projected Zombie cost (unit $u = 5.0861 \times 10^{-6}$ s B^{-2}, taken at $L_T = 2,000$) vs. BORA's net main-circuit folding cost. Est. Zombie = $u \cdot$ (corpus) $\cdot$ (regex set), a full document $\times$ regex cross product. Regex-set size is on-disk rule-file bytes for MAL/DNA; for DLP it is the pattern+keyword bytes fed to the circuit (the on-disk `.regex` doubles each sub-pattern for the bidirectional proximity form). The MAL (ClamAV) BORA cost 117.00 hr is the total over its 8 parallel jobs. The DLP (Enron) BORA cost 507.94 hr is the total over its 8 parallel jobs.

Dataset	Corpus	Regex set	Est. Zombie	BORA (net)	Speedup
MAL (ClamAV)	765.5 MB	11.6 MB	4.53×10^{10} (1436.9 yr)	4.21×10^5 (117.00 hr)	$\approx 107,661\times$
DNA (chr17)	83.3 MB	1.4 MB	5.99×10^8 (19.0 yr)	1.14×10^4 (3.16 hr)	$\approx 52,589\times$
DLP (Enron)	1.38 GB	213.2 KB	1.49×10^9 (47.3 yr)	1.83×10^6 (507.94 hr)	$\approx 816\times$

Table 11: End-to-end prover cost of Zombie [?], Reef [?], and BORA on the three datasets of Table 1. Zombie costs are projected from the unit cost $u = 5.0861 \times 10^{-6}$ s B^{-2} (Table 9). Values marked * are projected from Reef's cheapest measured per-run cost on DNA (7.33 s, the `proj_512k` bucket): the DNA floor as that cost per signature, and Reef MAL/DLP as that cost per signature per corpus byte (Table 8); the DNA cells pair Reef's as-implemented cost with that floor in parentheses. MAL and DLP BORA costs are net totals summed over 8 parallel jobs.

Dataset	Zombie	Reef	BORA	Speedup	
				vs Zombie	vs Reef
MAL	1436.9 yr	6.9 Myr*	117.00 hr	107,661×	518M×*
DNA	19.0 yr	300.9 d (2.3 d*)	3.16 hr	52,589×	2,284× (17.71×*)
DLP	47.3 yr	43,478 yr*	507.94 hr	816×	750,339×*